FACULTY WORKSHOP · WORKSHOP ONE

MACHINE LEARNING UNIVERSITY

Git + GitHub

for Faculty

aws

A 60-minute browser-only warm-up.
No installs. No terminal. No passwords.

AWS MLU Faculty Fellowship · Delaware State University · College of Agriculture, Science & Technology

YOUR INSTRUCTOR

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THE PROMISE

By the end of this hour, you will have...

01

Forked a real repository

Made your own copy of aws-dsu/my-first-repo in your GitHub account.

02

Made your first commit

Edited a file in the browser and saved a labeled snapshot of the change.

03

Tried github.dev

Opened a full code editor in a browser tab — no install required.

You will hold the wheel before you touch a clutch.

0–5 MIN · GROUND RULES

How this hour will work

Mics	Stay muted. We will not unmute the room of 600.
Chat	For clarifying questions only. Moderators will batch and pin answers.
Moderators	Three named moderators will spot-check your work and DM if you get stuck.
Recording	This session is being recorded with chapter markers. You can re-watch the fork demo later.
Pace	If you fall behind on the hands-on, that is fine. The recording will catch you up.

AGENDA

60 minutes at a glance

0–5	5–13	13–23	23–35	35–48	48–55	55–60
Welcome	**Git literacy**	**GitHub tour**	**Fork the repo**	**Edit & commit**	**github.dev bonus**	**Preview & wrap**

- Hands-on (you actually do something)
- Watch & learn (I drive)

If you only stay for one segment — stay for 23–48 minutes. That is when you fork and commit.

MACHINE LEARNING UNIVERSITY

aws

01

SECTION

Git Literacy

What git is. What GitHub is. Why educators care.

5–13 MIN · THE DEFINITION YOU CAN KEEP

Git is a time machine for files. GitHub is the cloud where time machines live.

By the end of this hour, you will own a time machine — and you will know how to point it at the AWS course materials.

Avoid the Dropbox / Google Drive / Word-Track-Changes analogies. They break within five minutes.

TWO THINGS, NOT ONE

Git is not GitHub. GitHub is not git.

git

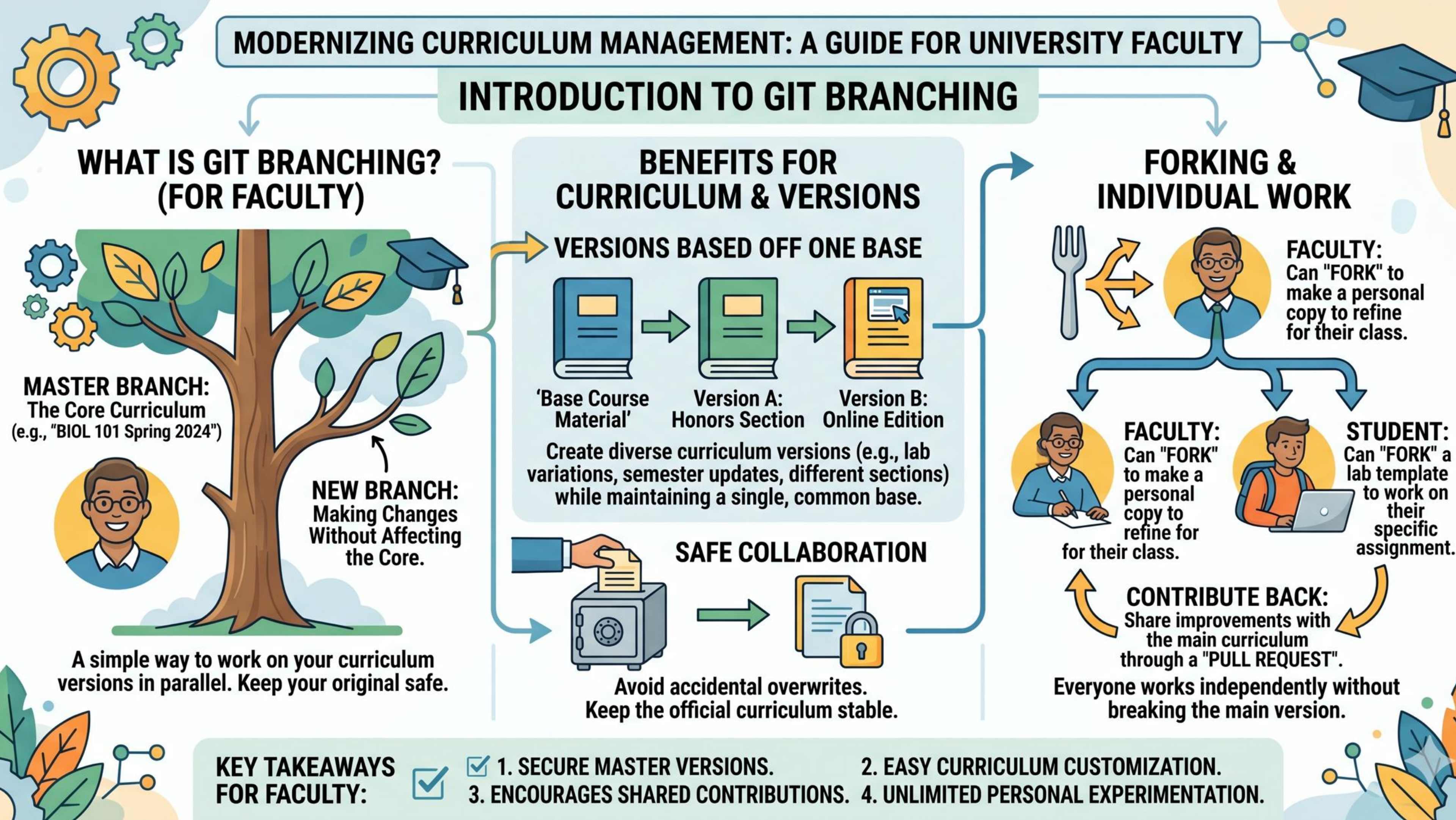
the tool

A tool that records every change to a folder of files. Any past version can be recovered. Any change can be explained — by author, by time, by message.

GitHub

the website

A website that hosts git-tracked folders so people can share them, see each other's changes, and collaborate. Today everything we do happens here.


MODERNIZING CURRICULUM MANAGEMENT: A GUIDE FOR UNIVERSITY FACULTY
INTRODUCTION TO GIT BRANCHING
WHAT IS GIT BRANCHING? (FOR FACULTY)
MASTER BRANCH:
The Core Curriculum
(e.g., "BIOL 101 Spring 2024")
NEW BRANCH:
Making Changes Without Affecting the Core.
A simple way to work on your curriculum versions in parallel. Keep your original safe.
BENEFITS FOR CURRICULUM & VERSIONS
VERSIONS BASED OFF ONE BASE
'Base Course Material'
Version A: Honors Section
Version B: Online Edition
Create diverse curriculum versions (e.g., lab variations, semester updates, different sections) while maintaining a single, common base.
SAFE COLLABORATION
Avoid accidental overwrites.
Keep the official curriculum stable.
FORKING & INDIVIDUAL WORK
FACULTY:
Can "FORK" to make a personal copy to refine for their class.
FACULTY:
Can "FORK" to make a personal copy to refine for for their class.
STUDENT:
Can "FORK" a lab template to work on their specific assignment.
CONTRIBUTE BACK:
Share improvements with the main curriculum through a "PULL REQUEST".
Everyone works independently without breaking the main version.
KEY TAKEAWAYS FOR FACULTY:
1. SECURE MASTER VERSIONS.
2. EASY CURRICULUM CUSTOMIZATION.
3. ENCOURAGES SHARED CONTRIBUTIONS.
4. UNLIMITED PERSONAL EXPERIMENTATION.

THE FULL VOCABULARY

Four verbs you will keep hearing

`clone`	Copy a repository from GitHub down to your computer or a cloud environment.	*= Fork (in the browser)*
`edit`	Change one or more files.	*= Pencil icon or github.dev*
`commit`	Save a labeled snapshot of your changes to the project history.	*= Commit changes button*
`sync`	Send your commits up to GitHub, or pull GitHub's commits down to your machine.	*= Sync in github.dev*

TODAY'S WORKFLOW

Fork → Edit → Commit

F

Fork

Click the Fork button.

Now you own a copy. The original is untouched.

E

Edit

Click the pencil icon.

Change the file in the web editor. Your fork only.

C

Commit

Click Commit changes.

Save a labeled snapshot. It is now in your history.

Fork ≈ Remix. The mental move you practice today is the same one you will practice in PartyRock next session.

WHY THIS HOUR IS WORTH IT

Five reasons educators learn git

1. **Distribute materials**

   Push course resources to students with one link — and update them in place.

2. **Customize official labs**

   Fork AWS or vendor materials and adapt them for your syllabus, keeping the link to the original.

3. **Pull updates cleanly**

   When AWS revises a lab, pull the updates into your fork without redoing your edits.

4. **Public teaching portfolio**

   Your repositories become a durable, citable record of your course design work.

5. **Collaborate with peers**

   Co-author labs and exercises with colleagues across institutions.

02

SECTION

GitHub Tour

I drive. You watch. We name the parts you will use today.

13–23 MIN · WHAT TO LOOK FOR

Anatomy of a GitHub repository

`Live on screen: github.com/aws-dsu/my-first-repo`

Organization	aws-dsu — the org that owns today's practice repo.
Repository	my-first-repo — the small, friendly target you will fork.
README	Renders on the front page. Where projects explain themselves.
File tree	Every file in the repo, in folders.
Commits tab	Every change ever made, with author and timestamp.
Code button	Green. Holds the URL you will later use for git clone.
Fork button	Top-right. The one you will click in five minutes.

Branches exist. You do not need to use them today. We will name them and move on.

03

SECTION

Hands-on #1 — Fork

Your turn. 12 minutes. Everyone has a fork by the end.

Fork aws-dsu/my-first-repo

```
github.com/aws-dsu/my-first-repo
```

1. Sign in to github.com.
2. Navigate to github.com/aws-dsu/my-first-repo.
3. Click the Fork button in the top-right.
4. Accept the defaults on the next page.
5. Paste your fork's URL into the chat as a check-in.

CHECK YOURSELF

What success looks like

BEFORE

```
github.com/aws-dsu/my-first-repo
```

AFTER

```
github.com/<your-username>/my-first-repo
```

Your username in the URL is the marker of ownership. Anything you change now changes only your copy.

04

SECTION

Hands-on #2 — Edit & Commit

First commit in the browser. No terminal. No install.

35–48 MIN · RECIPE 2

Edit a file. Commit the change.

1. On your fork, click README.md.
2. Click the pencil icon in the top-right of the file view.
3. Add a new line at the top of the file.
4. Scroll down. Type a commit message.
5. Confirm "Commit directly to main."
6. Click Commit changes. Refresh the Commits tab.

USE THESE

Add this line:

```
# My PartyRock app ideas
```

Commit message:

```
my first commit
```

Thematically primes the next session — PartyRock.

FIND YOUR COMMIT

Click the Commits tab.

Your commit is at the top of the list.

Your name

Author — that's you.

Your timestamp

When the change happened, to the second.

Your message

What you said about the change.

That is what git actually is — made visible.

05

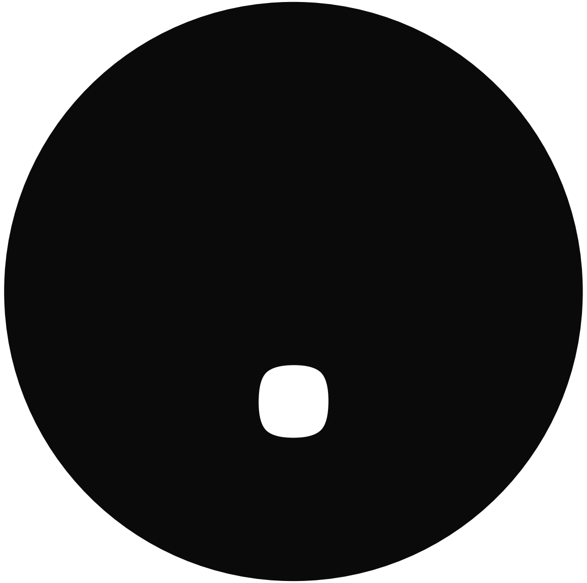
SECTION

Bonus — github.dev

A real editor. In a browser tab. No install.

48–55 MIN · THE TRICK

On any GitHub repo page…



…press the period key.

Or navigate directly to github.dev/<your-username>/my-first-repo.

WHAT JUST OPENED

A full editor — in a browser tab

- **File tree on the left** — Same as a desktop IDE: folders, files, click to open.
- **Open two files side by side** — Drag a tab to split the view. Useful for comparing notebooks.
- **Source Control panel** — The third icon on the left. Stage, commit, sync — all in the browser.
- **Sync back to github.com** — Click the sync icon. Refresh your repo page. The commit is there.

Many faculty use github.dev for course-prep editing even when they have a local setup. It is the fastest way to make small changes without leaving the browser.

06

SECTION

What Comes Next

The same loop — applied to the actual AWS course materials.

In your first lab, you will run this:

```
$ git clone https://github.com/aws-samples/aws-mlu-eep-generative-ai.git
```

the command *the AWS MLU course repository URL*

That is the FEC muscle, applied to the actual course repository.

You don't need to do it today. Workshop Studio will provide the terminal in your lab session.

BEFORE YOU LEAVE

Three rules of public-repo hygiene

01

Every commit is permanent

Even if you delete a file later, the commit history still contains it. Treat every commit as public the moment you click the button.

02

No identifiable student data, ever

No names, no grades, no IEP/504 information, no accommodation records, nothing FERPA-protected.

03

No institutional secrets

No internal budgets, personnel files, private policy drafts, or unpublished research data.

POST-SESSION

Three things to do before your first lab

1

Bookmark your fork URL

You will need it in Workshop Studio. Save it somewhere you will actually find it.

2

Make a second commit

Any file, any change. Reinforces the loop while it is fresh.

3

Join the async help channel

Post one question, even a small one. The channel is open for 72 hours.

EXIT TICKET

Before you go — drop these in chat

- [ ] My fork URL is ________________________
- [ ] The commit message on my first commit was ________________________
- [ ] One thing I will use my fork for in my teaching is ________________________
- [ ] One question I still have about git is ________________________

Or fill the follow-up form linked in the chat. Either is fine.

THAT'S THE WHOLE LOOP

Fork.
Edit.
Commit.

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Thank you. See you in the lab.